# Mathematical Logic

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# Contents

# 1 Mathematical Logic

## I Proof Theory

### i Deductive system, proof system, formal system, formal proof system, proof calculus, or deductive calculus

A deductive system, proof system, formal system, or formal proof system $\Gamma$ over a formal language $\mathcal{L}$ consists of axioms and rules of inference, where:

- An axiom is a formula in $\mathcal{L}$ assumed to be derived in $\Gamma$.
- An axiom schema is an expression with schema variables, each designated with a type of symbols in $\mathcal{L}$, that the expression obtained by replacing each schema variable with an arbitrary symbol of its type, called an application of the axiom schema, is assumed to be an axiom.
- A rule of inference in Hilbert style is in the form $\dfrac{P_1, P_2, \ldots, P_n}{Q}$ where $P_1, P_2, \ldots, P_n, Q$ are expressions sharing the same set of metaformulas such that any expressions obtained by replacing each metaformula with an arbitrary formula in $\mathcal{L}$ are formulas in $\mathcal{L}$ and the expressions obtained by such replacement of $P_1, P_2, \ldots, P_n$ are assumed to derive the expression obtained by such replacement of $Q$, in which $P_1, P_2, \ldots, P_n$ and their replacements are called preimises and $Q$ and their replacements called conclusion. Other styles exist and have the same expressive power as Hilbert style for propositional logic and predicate logic.

### ii (Formal) derivation or proof

Given proof system $\Gamma$ over language $\mathcal{L}$ and formulas $S_1, S_2, \ldots S_n, T \in \mathcal{L}$, a (formal) derivation or proof from $S_1, S_2, \ldots S_n$ to $T$ is a sequence of formulas $\langle F_i \rangle_{i=1}^m$ such that each $F_i$ is either

- an axiom of $\Gamma$,
- an element of $\{S_1, S_2, \ldots S_n\}$, or
- can be derived from some of the earlier formulas in the sequence using a rule of inference,

and that $F_m = T$.

$S_1, S_2, \ldots S_n$ are called premises, $T$ is called conclusion, and this process is called argumentation.

That there exists a derivation from $S_1, S_2, \ldots S_n$ to $T$ is denoted as $S_1, S_2, \ldots S_n \vdash T$ and called $S_1, S_2, \ldots S_n$ derive or prove $T$.

Thus, for any axiom $\alpha$, $\vdash \alpha$.

Any formula $\alpha$ such that $\vdash \alpha$ is called derived or a theorem and is also denoted as $\Gamma \vdash \alpha$.

### iii Finitarity

A rule of inference is finitary iff its premises are finite.

The rules of inference and sequences in formal proofs are required to be finite for finitary deductive systems.

### iv Soundness

A deductive system is sound if every formula that can be derived from the system is a tautology in the language.

### v Syntactic consequence

For formulas $S_1, S_2, \dots S_n, T$ in a language $\mathcal{L}$, $T$ is a syntactic consequence of $S_1, S_2, \dots S_n$, denoted as $S_1, S_2, \dots S_n \vdash T$, iff there exists a sound proof system over $\mathcal{L}$ in which $S_1, S_2, \dots S_n \vdash T$.

### vi Independent formula

A formula $P$ such that neither $\vdash P$ nor $\vdash \neg P$.

### vii (Syntactical) completeness

A deductive system is (syntactically) complete if for every formula in the language, either it or its negation can be derived from the system.

### viii Semantical completeness

A deductive system is semantically complete if every tautology in the language can be derived from the system.

### ix Interpretability

If a deductive system $S$ over language $L$ and a deductive system $T$ over language $M$ are such that there exists a function $H : L \to M$ such that $S \vdash l \implies T \vdash H(l)$, then we say $T$ intepretes $S$, $S$ is weaker than $T$, or $T$ is stronger than $S$.

### x Computably enumerable (c.e.), recursively enumerable (r.e.), semidecidable, partially decidable, listable, provable or Turing-recognizable deductive system

A deductive system is computably enumerable (c.e.), recursively enumerable (r.e.), semidecidable, partially decidable, listable, provable or Turing-recognizable iff the set of all axioms of it is computably enumerable.

If a deductive system is computably enumerable and its rules of inferences are finitary, it is finitary.

# II Model Theory

## i Signature

A signature $\sigma = (\mathcal{P}, \mathcal{F})$ is a pair of indexed sets,

- a set of sets of predicate symbols (also called relation symbols) $\mathcal{P} = \{P_n \mid n \in \mathbb{N}_0\}$ with $P_n$ being a possibly infinite set of predicate symbols of arity $n$,
- a set of sets of function symbols $\mathcal{F} = \{F_n \mid n \in \mathbb{N}_0\}$ with $F_n$ being a possibly infinite set of function symbols of arity $n$, with any element of $F_0$ called a constant.

## ii (Logical or model-theoretic) structure

A (logical or model-theoretic) structure can be defined with a triple $(A, \sigma, I)$ of a set called domain, universe, carrier (set), underlying set, domain of discourse, or universe of discourse, $A$, denoted as $|\mathcal{A}|$ or simply $\mathcal{A}$, a signature $\sigma$, and an interpretation function $I$ that, maps for all $n \in \mathbb{N}_0$, maps each $f \in F_n$ to an $n$-ary operation $I(f)$ on $A$, called interpretation of $f$ and often denoted as $f$, and for all $n \in \mathbb{N}_0$, maps each $p \in P_n$ to a subset $I(p)$ of $A^n$, called interpretation of $p$ and often denoted as $p$, in which $A^0 = \{\emptyset\}$.

A structure with signature $\sigma$ is called a $\sigma$-structure. A structure with signature $\sigma$ of a language $L$ is called a $L$-structure.

## iii Substructure

A structure $\mathcal{B} = (B, \sigma_B, I_B)$ is a substructure of a structure $\mathcal{A} = (A, \sigma_A, I_A)$, that is, $\mathcal{A}$ is an extension of $\mathcal{B}$, denoted as $\mathcal{B} \subseteq \mathcal{A}$, iff

- $B \subseteq A$,
- $\sigma_B = \sigma_A$,
- for every $n \in \mathbb{N}_0$, for every function symbol $f \in F_n$, $I_B(f) = I_A(f) \restriction B^n$, and
- for every $n \in \mathbb{N}_0$, for every relation symbol $p \in P_n$, $I_B(p) = I_A(p) \cap B^n$.

## iv Elementary substructure

A structure $\mathcal{B} = (B, \sigma_B, I_B)$ is an elementary substructure of a structure $\mathcal{A} = (A, \sigma_A, I_A)$, that is, $\mathcal{A}$ is an elementary extension of $\mathcal{B}$, denoted as $\mathcal{B} \prec \mathcal{A}$, iff, with the language of $\sigma_B$ denoted as $\mathcal{L}$,

- $\mathcal{B} \subseteq \mathcal{A}$,
- for every $n$-ary formula $\varphi \in \mathcal{L}$, for every $a \in B^n$, $\mathcal{B} \vdash \varphi[a] \iff \mathcal{A} \vdash \varphi[b]$.

## v Closed subset

For a structure $\mathcal{A} = (A, \sigma, I)$, a subset $B \subseteq A$ is called closed iff, for all $n \in \mathbb{N}_0$, for any $f \in F_n$ and $b \in B^n$, $I(f)(b) \in B$.

For any subset $B$ of $A$, there exists a smallest closed subset that is a superset of $B$, called the hull of $B$ or the closed subset derived by $B$, denoted as $\langle B \rangle_{\mathcal{A}}$.

### vi Cardinality

The cardinality of a structure is defined as the cardinality of its domain.

### vii Homomorphism

A ($\sigma$−)homomorphism from a structure $\mathcal{A} = (A, \sigma, I_A)$ to a structure $\mathcal{B} = (B, \sigma, I_B)$ is a map $h\colon A \to B$ such that,

- for all $n \in \mathbb{N}_0$, for any $f \in F_n$ and $a_1, a_2, \ldots, a_n \in A^n$,

$$h(I_A(f)(a_1, a_2, \ldots, a_n)) = I_B(f)(h(a_1), h(a_2), \ldots, h(a_n)),$$

- for all $n \in \mathbb{N}_0$, for any $p \in P_n$ and $a_1, a_2, \ldots, a_n \in A^n$,

$$(a_1, a_2, \ldots, a_n) \in I_A(p) \implies (h(a_1), h(a_2), \ldots, h(a_n)) \in I_B(p).$$

For any $a \in A^n$ with $a = (a_1, a_2, \ldots, a_n)$, $(h(a_1), h(a_2), \ldots, h(a_n))$ is denoted as $h(a)$.

### viii Strong homomorphism

A strong ($\sigma$−)homomorphism from a structure $\mathcal{A} = (A, \sigma, I_A)$ to a structure $\mathcal{B} = (B, \sigma, I_B$ is a map $h\colon A \to B$ such that,

- for all $n \in \mathbb{N}_0$, for any $f \in F_n$ and $a_1, a_2, \ldots, a_n \in A^n$,

$$h(I_A(f)(a_1, a_2, \ldots, a_n)) = I_B(f)(h(a_1), h(a_2), \ldots, h(a_n)),$$

- for all $n \in \mathbb{N}_0$, for any $p \in P_n$ and $a_1, a_2, \ldots, a_n \in A^n$,

$$(a_1, a_2, \ldots, a_n) \in I_A(p) \iff (h(a_1), h(a_2), \ldots, h(a_n)) \in I_B(p).$$

### ix Embedding

A ($\sigma$−)embedding from a structure $\mathcal{A} = (A, \sigma, I_A)$ to a structure $\mathcal{B} = (B, \sigma, I_B$ is a strong homomorphisms $h$ from $\mathcal{A}$ to $\mathcal{B}$ that is injective.

### x Isomorphism

A ($\sigma$−)isomorphism from a structure $\mathcal{A} = (A, \sigma, I_A)$ to a structure $\mathcal{B} = (B, \sigma, I_B$ is a strong homomorphisms $h$ from $\mathcal{A}$ to $\mathcal{B}$ that is bijective.

### xi Isomorphism-invariance

An $n$-free variable formula $\varphi$ is isomorphism-invariant between two models $\mathcal{M}, \mathcal{N}$ with domains $M, N$ iff for every isomorphism $h$ between $\mathcal{M}, \mathcal{N}$, for any $a \in M^n$:

$$\mathcal{M} \vdash \varphi[a] \iff \mathcal{N} \vdash \varphi[h(a)].$$

### xii Definability

Let $L$ be a first-order language, $\mathcal{M}$ be a $L$-structure with domain $M$, $X$ be a fixed subset of $M$, and $m$ be a positive integer.

- A set $A \subseteq M^m$ is definable in $\mathcal{M}$ with parameters from $X$ iff there exists a formula $\varphi$ of $m + n$ free variables and $b_1, b_2, \dots, b_n \in X$ such that $\forall a_1 \in M \forall a_2 \in M \dots \forall a_m \in M$, $(a_1, a_2 \dots, a_m) \in A$ iff $\mathcal{M} \vDash \varphi[a_1, a_2, \dots, a_m, b_1, b_2, \dots, b_n]$, and $\varphi[a_1, a_2, \dots, a_m, b_1, b_2, \dots, b_n]$ with parameters $b_1, b_2, \dots, b_n$ is called defining formula.
- A set is definable in $\mathcal{M}$ without parameter iff it is definable in $\mathcal{M}$ with parameters from $\varnothing$.
- A function is definable in $\mathcal{M}$ (with some parameters) iff its graph is definable in $\mathcal{M}$ (with those parameters).
- An element $a \in M^m$ is definable in $\mathcal{M}$ (with some parameters) iff $\{a\}$ is definable in $\mathcal{M}$ (with those parameters).
- Set of definable sets: $\mathrm{Def}(\mathcal{M})$ is the union for all $m \in \mathbb{N}$ of the set of subsets of $M^m$ that is definable in $\mathcal{M}$ (with any parameter or without parameter).

## III Propositional logic (PL), statement logic, sentential logic, propositional calculus, statement calculus, sentential calculus, or zeroth-order logic

### i Syntax

Given

- a set of connectives, logical connectives, logical operators, truth-functional connectives, truth-functors, or propositional connectives $\mathcal{O} = \{\neg, \wedge, \vee\}$ with unary negation $\neg$, binary conjunction $\wedge$, and binary disjunction $\vee$, (or any functionally complete set of logical operators in Boolean algebra with related definitions changed accordingly), and
- a set of atomic propositions, atomic formulas, atomic sentences, atoms, proposition letters, sentence letters, propositional variables, or variables $\mathcal{P}$ with $\mathcal{P} \cap \mathcal{O} = \varnothing$ and $\mathcal{O} \cup \mathcal{P}$ be the alphabet of $\mathcal{L}$,

the formal language $\mathcal{L}$ of propositional logic is defined by:

- $\mathcal{P} \subseteq \mathcal{L}$,
- $\neg P \in \mathcal{L}$ if $P \in \mathcal{L}$,
- $(P \wedge Q) \in \mathcal{L}$ if $P, Q \in \mathcal{L}$,
- $(P \vee Q) \in \mathcal{L}$ if $P, Q \in \mathcal{L}$, and
- nothing else is in $\mathcal{L}$.

Any $w \in \mathcal{L}$ is called a well-formed formula (WFF or wff), or simply formula.

## ii Semantics

For a given propositional logic, its interpretation (or valuation) is defined as an interpretation (or valuation) function of its set of atomic propositions.

For a given set of atomic propositions $\mathcal{P}$, an interpretation (or valuation) function $\mathcal{I}$ maps every formula in $\mathcal{P}$ to a semantic value in $\{1,0\}$ (with 1 meaning true and 0 meaning false), denoted, for each $P \in \mathcal{P}$, as

$$\mathcal{I}(P) = B.$$

For a given propositional logic with formal language $\mathcal{L}$ and interpretation $\mathcal{I}$, the interpretation function $\mathscr{I}$ over $\mathcal{L}$ is recursively defined as

- for any $P \in \mathcal{P}$, $\mathscr{I}(P) = \mathcal{I}(P)$,
- for any $P \in \mathcal{L}$, $\mathscr{I}(\neg P) = \neg\mathscr{I}(P)$,
- for any $P, Q \in \mathcal{L}$, $\mathscr{I}(P \wedge Q) = \mathscr{I}(P) \wedge \mathscr{I}(Q)$, and
- for any $P, Q \in \mathcal{L}$, $\mathscr{I}(P \vee Q) = \mathscr{I}(P) \vee \mathscr{I}(Q)$.

For any $P \in \mathcal{L}$, we write $\mathcal{I} \vDash P$ if $\mathscr{I}(P) = 1$ and $\mathcal{I} \nvDash P$ if $\mathscr{I}(P) = 0$.

Alternatively, with structure, as expressed in the language of first-order logic, we can define each atomic formula in propositional logic as a nullary predicate symbol, that there is no function symbol, and that the interpretation structure maps each nullary predicate symbol to a subset of $\{\varnothing\}$, that is, either $\{\varnothing\}$ or $\varnothing$, and with $\varnothing \in \{\varnothing\}$ and $\varnothing \notin \varnothing$, an atomic formula mapped to $\{\varnothing\}$ is true and an atomic formula mapped to $\varnothing$ is false.

## iii Semantic consequence

Given formulas $S_1, S_2, \ldots S_n, T$, $S_1, S_2, \ldots S_n \vDash T$, called that $T$ is a semantic consequence of $S_1, S_2, \ldots S_n$, means that for any interpretation function $\mathcal{I}$, if $\mathcal{I} \vDash S_1, \mathcal{I} \vDash S_2, \ldots, \mathcal{I} \vDash S_n$, then $\mathcal{I} \vDash T$.

## iv Equivalence of syntactic and semantic consequences

For any formulas $S_1, S_2, \ldots S_n, T$,

$$S_1, S_2, \ldots S_n \vdash T \iff S_1, S_2, \ldots S_n \vDash T.$$

Thus syntactic and semantic consequences are called (logical) consequences.

## v Classification of formulas

A formula $P$ is called (logically) valid or a tautology, denoted as $\top$, iff for any interpretation function $\mathcal{I}$, $\mathcal{I} \vDash P$.

A formula $P$ is called (logically) invalid or a contradiction, denoted as $\bot$, iff for any interpretation function $\mathcal{I}$, $\mathcal{I} \nvDash P$.

A formula $P$ is called contingent iff it is neither a tautology nor a contradiction.

A formula $P$ is called satisfiable iff it is not a contradiction.

### vi Typical rules of inference

For all equations that hold in Boolean algebra, with each Boolean variable replaced with a metaformula, LHS as numerator and RHS as denominator.

## IV First-order logic (FOL), predicate logic, predicate calculus, or quantificational logic

Proportional logic and predicate logic are collectively called classical logic.

### i Syntax

Given logic symbols,

- (either one of) the quantifier symbols $\forall$ for universal quantification and $\exists$ for existential quantification,
- a set of logical connectives $\mathcal{O} = \{\neg, \wedge, \vee, \implies\}$ with unary negation $\neg$, binary conjunction $\wedge$, binary disjunction $\vee$, and binary material implication $\implies$, (or any functionally complete set of logical operators in Boolean algebra with related definitions changed accordingly),
- a set of punctuation symbols $\mathcal{N}$ containing $($, $)$, $,$ and maybe other symbols, which do not carry meaning themselves but to help parse and read the language,
- an infinite set of variables $\mathcal{V}$,
- an equality symbol $=$ for first-order logic with equality,

and a signature $\sigma = (\mathcal{P}, \mathcal{F})$ with the equality symbol $= \in P_2$ for first-order logic with equality, the set of terms $\mathcal{T}$ is defined by:

- $\mathcal{V} \subseteq \mathcal{T}$, and
- for all $n \in \mathbb{N}_0$, for all $t \in \mathcal{T}^n$, for all $f \in F_n$, $f(t) \in \mathcal{T}$,

the set of atomic formulas $\mathcal{C}$ is defined as

$$\mathcal{C} = \bigcup_{n \in \mathbb{N}_0} \{p(t) \mid t \in \mathcal{T}^n \wedge p \in P_n\},$$

and the formal language $\mathcal{L}$ of first-order logic is defined by:

- $\mathcal{C} \subseteq \mathcal{L}$
- $\neg P \in \mathcal{L}$ if $P \in \mathcal{L}$,
- $(P \wedge Q) \in \mathcal{L}$ if $P, Q \in \mathcal{L}$,
- $(P \vee Q) \in \mathcal{L}$ if $P, Q \in \mathcal{L}$,
- $(P \implies Q) \in \mathcal{L}$ if $P, Q \in \mathcal{L}$, and
- $\forall x P, \exists x P \in \mathcal{L}$ if $P \in \mathcal{L}$ and $x \in \mathcal{V}$,

- nothing else is in $\mathcal{L}$.

Any $w \in \mathcal{L}$ is called a well-formed formula (WFF or wff) or simply formula.

A variable in a formula is either free or bounded, which are defined as follows for formulas $P$ and $Q$:

- A variable in an atomic formula is free.
- A variable is free in $\neg P$ if it is free in formula $P$.
- A variable is free in $P \wedge Q$ if it is free in formula $P$ or $Q$.
- A variable is free in $P \vee Q$ if it is free in formula $P$ or $Q$.
- A variable is free in $P \implies Q$ if it is free in formula $P$ or $Q$.
- A variable $x$ is free in $\forall y P$ and $\exists y P$ if $x$ is not the same symbol as $y$ and $x$ is free in formula $P$.
- No other variable in a formula is free.

A formula with no free variable is called a sentence or a closed formula.

A formula $P$ with free variables $x_1, x_2, \ldots, x_n$ is denoted as $P[x_1, x_2, \ldots, x_n]$ or $P(x_1, x_2, \ldots, x_n)$.

The cardinalities of $\sigma$ and $\mathcal{L}$ are both defined as the cardinality of $(\bigcup_{u \in \mathcal{P}} u) \cup (\bigcup_{u \in \mathcal{F}} u)$.

### ii Semantics

For a given first-order logic, its interpretation (or valuation) is defined as $\mathcal{I} = \mathcal{A}, s$ with an interpretation (or valuation) structure $\mathcal{A} = (A, \sigma, I)$ of the same signature as the first-order logic with the constraint for first-order logic with equality that

$$I(=) = \{(a, a) \mid a \in A\}$$

and a variable assignment function $s$ from the variables $\mathcal{V}$ to $A$.

For any formula $P$, $\mathcal{A}, s \vDash P$ denotes that $P$ is true in $\mathcal{A}$ under assignment $s$, and $\mathcal{A}, s \nvDash P$, denotes that $P$ is false in $\mathcal{A}$ under assignment $s$.

For a given first-order logic with formal language $\mathcal{L}$ and interpretation $\mathcal{A}, s$,

- for any term $t$, $[t]$ or $[t]_{\mathcal{A},s}$ denotes $t$ with any function symbols $f$ replaced with $I(f)$ and any variable $x$ replaced with $s(x)$,
- for any atomic formula $p(t_1, t_2, \ldots, t_n)$ for some predicate symbol $p \in P_n$, $\mathcal{A}, s \vDash p(t)$ iff $([t_1], [t_2], \ldots, [t_n]) \in I(p)$, and
- T-schema or truth schema: with $\mathscr{I}$ mapping formula $P$ to 1 if $\mathcal{A}, s \vDash P$ and to 0 if $\mathcal{A}, s \nvDash P$,
  - for any $P \in \mathcal{L}$, $\mathscr{I}(\neg P) = \neg \mathscr{I}(P)$,
  - for any $P, Q \in \mathcal{L}$, $\mathscr{I}(P \wedge Q) = \mathscr{I}(P) \wedge \mathscr{I}(Q)$,
  - for any $P, Q \in \mathcal{L}$, $\mathscr{I}(P \vee Q) = \mathscr{I}(P) \vee \mathscr{I}(Q)$,

- for any $P, Q \in \mathcal{L}$, $\mathscr{I}(P \implies Q) = \mathscr{I}(P) \implies \mathscr{I}(Q)$,
- $\mathcal{A}, s \vDash \forall x P$ iff $\forall a \in A$, replacing $x$ with $a$ makes $\mathcal{A}, s \vDash P$, that is, $\forall x P = \bigwedge_{a \in A} P[x \mapsto a]$, and
- $\mathcal{A}, s \vDash \exists x P$ iff $\exists a \in A$, replacing $x$ with $a$ makes $\mathcal{A}, s \vDash P$, that is, $\exists x P = \bigvee_{a \in A} P[x \mapsto a]$.

For sentences, $\mathcal{A}, s$ can be denoted as $\mathcal{A}$ since there is no free variable and thus $s$ is irrelevant.

### iii Semantic consequence

Given formulas $S_1, S_2, \ldots S_n, T$, $S_1, S_2, \ldots S_n \vDash T$, called that $T$ is a semantic consequence of $S_1, S_2, \ldots S_n$, means that for any interpretation $\mathcal{A}, s$, if $\mathcal{A}, s \vDash S_1, \mathcal{A}, s \vDash S_2, \ldots, \mathcal{A}, s \vDash S_n$, then $\mathcal{A}, s \vDash T$.

### iv Equivalence of syntactic and semantic consequences

For any formulas $S_1, S_2, \ldots S_n, T$,

$$S_1, S_2, \ldots S_n \vdash T \iff S_1, S_2, \ldots S_n \vDash T.$$

Thus syntactic and semantic consequences are called (logical) consequences.

The $\Leftarrow$ direction is the Gödel's completeness theorem and the $\Rightarrow$ direction is from the Henkin's model existence theorem.

### v Classification of formulas

A formula $P$ is called (logically) valid or a tautology, denoted as $\top$, iff for any interpretation $\mathcal{A}, s$, $\mathcal{A}, s \vDash P$.

A formula $P$ is called (logically) invalid or a contradiction, denoted as $\perp$, iff for any interpretation $\mathcal{A}, s$, $\mathcal{A}, s \nvDash P$.

A formula $P$ is called contingent iff it is neither a tautology nor a contradiction.

A formula is satisfiable iff it is not a contradiction.

### vi Typical rules of inference

- For all equations that hold in Boolean algebra, with each Boolean variable replaced with a metaformula, LHS as numerator and RHS as denominator,
- for any formula $P$ with one free variable, for any variable $a$,
$$\frac{\forall x P(x)}{P(a)}$$
(universal instantiation), and
- for any formula $P$ with one free variable, for any variable $a$,
$$\frac{P(a)}{\exists x P(x)}$$
(existential generalization).

# V (Formal) (Deductive) Theory

## i (Formal) (deductive) theory

A (formal) (deductive) theory $T$ over a formal language $\mathcal{L}$ is a set of formulas in $\mathcal{L}$ that is closed under $\vdash$. Each $P \in T$ is called a theorem.

## ii Deductive system

For a theory $T$ over language $\mathcal{L}$, a deductive system that $T = \{P \in \mathcal{L} \mid \vdash P\}$ is called to derive $T$ and is not distinguished with $T$ in terms of properties.

## iii (Syntactical) consistency

A theory $T$ is (syntactically) consistent iff there is no formula $P$ such that $P \in T$ and $\neg P \in T$.

## iv Satisfiability

A theory $T$ is satisfiable iff there exists an interpretation $\mathcal{I}$ such that for every $P \in T$, $\mathcal{I} \vDash P$, and $T$ is said to be satisfied by $\mathcal{I}$ or has an interpretation $\mathcal{I}$.

## v Complete (consistent) theory

A complete (consistent) theory $T$ is a consistent theory such that for every formula $P$ in its language, either $T \vdash P$ or $T \cup \{P\}$ is inconsistent.

## vi Subtheory

If a theory $S$ and a theory $T$ is such that $S \subseteq T$, then $S$ is called a subtheory of $T$, $T$ is called a supertheory or an extension of $S$.

## vii Effectively axiomatizable, computably axiomatizable, or recursively axiomatizable

A theory $T$ over a language $\mathcal{L}$ is effectively axiomatizable, computably axiomatizable, or recursively axiomatizable if there exists a deductive system with recursively enumerable set of axioms such that $T = \{P \in \mathcal{L} \mid \vdash P\}$, that is, there exists a finitary deductive system such that $T = \{P \in \mathcal{L} \mid \vdash P\}$ and there exists a recursively enumerable set of formal proofs that derive all $P \in T$.

## viii Independent formula

For any independent formula $P$ of a sound deductive system that generate a satisfiable theory $T$, $T \cup \{P\}$ and $T \cup \{\neg P\}$ are both satisfiable.

## ix (Complete) theory of interpretation

The (complete) theory of an interpretation $\mathcal{A}$ is all formulas that are satisfied by $\mathcal{A}$, denoted as $\mathrm{Th}(\mathcal{A})$.

The theory of a class $K$ of interpretations is all formulas that are satisfied by all $k \in K$, denoted as $\text{Th}(K)$.

# VI First-order theory

## i First-order theory

A (formal) (deductive) first-order theory $T$ over a formal language $\mathcal{L}$ is a set of sentences in $\mathcal{L}$ that is closed under $\vdash$.

## ii First-order theory with identity

A first-order theory is with identiy if its language has the equality symbol $=$, and the set of theorems of the theory contains all applications of the following axiom schemas:

- Reflexivity: For any variable $x$, $x = x$.
- Leibniz's law or substitution: For any $n \in \mathbb{N}_0$, for any formula with any $n$ free variables $x_1, x_2, \dots, x_n$, any $i \in \mathbb{N} \wedge i \leq n$, and any variable $x$,

$$(x_i = x) \implies (P(x_1, x_2, \dots, x_i, \dots, x_n) \implies P(x_1, x_2, \dots, x_{i-1}, x, x_{i+1}, \dots, x_n)).$$

## iii Equivalence of consistency and satisfiability

A theory over proportional or first-order language is consistent iff it is satisfiable.

The $\Leftarrow$ direction is from the Gödel's completeness theorem and the $\Rightarrow$ direction is the Henkin's model existence theorem.

## iv Elementary class or axiomatizable class

The elementary class or axiomatizable class of a first-order theory $T$ is the class of all structures that satisfies $T$.

## v Downward Löwenheim–Skolem Theorem

If a first-order theory with signature $\sigma$ has an infinite model $\mathcal{M}$ of some cardinality $\kappa \geq \aleph_0$, then for every $\mu$ such that $\max(|\sigma|, \aleph_0) \leq \mu \leq \kappa$, it has model of cardinality $\mu$ that is an elementary substructures of $\mathcal{M}$.

## vi Upward Löwenheim–Skolem Theorem

If a first-order theory with signature $\sigma$ has an infinite model $\mathcal{M}$ of some cardinality $\kappa \geq \aleph_0$, then for every $\mu \geq \kappa$, it has model of cardinality $\mu$ that is an elementary extension of $\mathcal{M}$.

## vii Compactness theorem

If a first-order theory $T$ is such that all its finite subsets have models, then $T$ has model.

## VII Gödel's incompleteness theorems

### i Gödel's first incompleteness theorem

Any recursively enumerable consistent deductive system that can interprets the Robinson arithmetic is syntactically incomplete.

### ii Gödel's second incompleteness theorem

Any recursively enumerable consistent deductive system that can interprets the Robinson arithmetic can not prove its consistency.

## VIII Abstract logic

### i Abstract logic

Let $L$ be a formal language closed under logical connectives and with functionally complete set of logical connectives, and $\mathscr{M}$ be all models (possibly designated to be allowed) for it. The ordered pair $(L, \mathscr{M})$ is called an abstract logic.

### ii Isomorphism-invariant

An abstract logic, $(L, \mathscr{M})$, is isomorphism-invariant iff for every two models $\mathcal{M}, \mathcal{N} \in \mathscr{M}$, for every sentence $\varphi \in L$, $\varphi$ is isomorphism-invariant between $\mathcal{M}$ and $\mathcal{N}$.

### iii Expressive power

All isomorphism-invariant abstract logics form a class $\mathbb{L}$ with partial order $\leq$, called weaker than or less expressive than, equivalence relation $=$, and strict partial order $<$, called strictly weaker than or strictly less expressive than, of expressive power defined, for any $(L_1, \mathscr{M}_1), (L_2, \mathscr{M}_2) \in \mathbb{L}$, with the $\mathscr{N}_1$ and $\mathscr{N}_2$ defined such that all elements of $\mathscr{N}_1 \cup \mathscr{N}_2$ have the same signature $\sigma$ and domain $N$, that for any $\mathcal{M} \in \mathscr{M}_1$, there exists at least one $\mathcal{N} \in \mathscr{N}_1$ such that there exists at least one isomorphism between $\mathcal{M}$ and $\mathcal{N}$, and that for any $\mathcal{M} \in \mathscr{M}_2$, there exists at least one $\mathcal{N} \in \mathscr{N}_2$ such that there exists at least one isomorphism between $\mathcal{M}$ and $\mathcal{N}$, as:

$$(L_1, \mathscr{M}_1) \leq (L_2, \mathscr{M}_2) \iff \bigcup_{\mathcal{N} \in \mathscr{N}_1} \mathrm{Def}(\mathcal{N}) \subseteq \bigcup_{\mathcal{N} \in \mathscr{N}_2} \mathrm{Def}(\mathcal{N}),$$

$$(L_1, \mathscr{M}_1) = (L_2, \mathscr{M}_2) \iff (L_1, \mathscr{M}_1) \leq (L_2, \mathscr{M}_2) \land (L_2, \mathscr{M}_2) \leq (L_1, \mathscr{M}_1).$$

$$(L_1, \mathscr{M}_1) < (L_2, \mathscr{M}_2) \iff (L_1, \mathscr{M}_1) \leq (L_2, \mathscr{M}_2) \land \neg(L_2, \mathscr{M}_2) \leq (L_1, \mathscr{M}_1).$$

### iv Compactness property

For an abstract logic $(L, \mathscr{M})$, if for every theory $T$ over its language, that all its finite subsets have model in $\mathscr{M}$ implies that $T$ has model in $\mathscr{M}$, then the abstract logic has compactness property.

Compactness property implies that for any theory has model in $\mathscr{M}$, there exists a sound finitary deductive system that derive it.

That for any theory of a model in $\mathscr{M}$, there exists a sound, finitary and semantically complete deductive system that derive it, implies compactness property.

### v Downward Löwenheim–Skolem property

For an abstract logic $(L, \mathscr{M})$ with signature $\sigma$, if for every theory $T$ over $L$ that has an infinite model $\mathcal{M} \in \mathscr{M}$ of some cardinality $\kappa \geq \aleph_0$, for every $\mu$ such that $\max(|\sigma|, \aleph_0) \leq \mu \leq \kappa$, it has model of cardinality $\mu$ that is an elementary substructures of $\mathcal{M} \in \mathscr{M}$, then the abstract logic is called to have downward Löwenheim–Skolem property.

### vi Upward Löwenheim–Skolem Theorem

For an abstract logic $(L, \mathscr{M})$ , if for every theory $T$ over $L$ that has an infinite model $\mathcal{M} \in \mathscr{M}$ of some cardinality $\kappa \geq \aleph_0$, then for every $\mu \geq \kappa$, it has model of cardinality $\mu$ that is an elementary extension of $\mathcal{M} \in \mathscr{M}$.

### vii Lindström's theorem

Any isomorphism-invariant abstract logic extending first-order logic with standard semantics that has compact property and downward Löwenheim–Skolem property must not has strictly more expressive power than first-order logic with standard semantics.